CSIS-250-GC: INTRODUCTION TO PYTHON PROGRAMMING

History

1. Oct 15, 2025 by lori.jones

Viewing: CSIS-250-GC : Introduction to Python Programming
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General Information

College
Grossmont

Subject Code (CB00)
Computer Sci Info Systems-GC (CSIS)

Course Number (CB20)
250

Long Title (CB01)
Introduction to Python Programming

Short Title
Intro to Python Programming

Department (CB01, CB19)
Computer Science & Systems (GCSIS)

Division
Career & Technical Ed/Workforce Dev

Effective Term
2018-2019

Catalog Description
This is an introductory course in Python programming. Topics covered include basic language syntax, functions, control flow, looping, data types, strings, lists, dictionaries, exception handling, software tools and libraries, and functional vs object oriented programming concepts. This course is intended for persons with a prior background in any programming language.

Instructional Methods
Lecture and Lab (LEC LAB)

Total Units
4

Lecture Hours
3.00

Lab Hours
3.00

Lecture Hours

Total Contact Hours - Min.
48

Total Contact Hours - Max.
54

Outside-of-Class Hours Min.
96

Outside-of-Class Hours Max.
108

Total Student Learning Hours Min.
144
Total Student Learning Hours Max
162

Lab Hours

Total Contact Hours Min.
48
Total Contact Hours Max.
54

Outside-of-Class Hours Min.
0
Outside-of-Class Hours Max.
0

Total Student Learning Hours Min.
48
Total Student Learning Hours Max.
54

Is this a computer-assisted lab?
No

Is the course graded pass/no-pass only?
No

Special Class Status (CB13)
N - Course is not a special class.

Recommended Preparation
"C" grade or higher or "Pass" in CSIS 119 or equivalent.

Course Details

TOP Code (CB03)
Computer Information Systems (070200)

Credit Status (CB04)
Credit - Degree Applicable

Noncredit-Category (CB22)
Y: Credit Course

SAM Code (CB09)
D: Possibly Occupational

Funding-Agency-Category (CB23)
N: Not VATEA Funding

Program Status (CB24)
Program Applicable

Curricular Notice

Does this proposal affect the other college?
No

Indicate any additional departments that are affected by this course due to curricular impact and/or duplication.

Proposal Details

Proposal rationale
Conversion of COR into CIM

Entrance Skills

Content Review Documentation

Exit Skills

Content Review Documentation

General Education

Meets general education requirements for:

Associate Degree (Local)
No

CSU
No

IGETC
No

Cal-GETC
No

Transferability (CB05)
Transfers to both UC/CSU

Course Alignment Verification

Alignment Criteria

Course Outline of Record

Course Content

1. Historical survey of Python and their place among other programming languages.
2. Portability between platforms.
3. Python program structure and language syntax.
4. Decisions (branching) in Python.
5. Iteration in Python.
6. Python operators.
7. Data structures including standard data types and abstract data types.
8. Functions and modules.
9. Input/Output.
10. Algorithms and Problem Solving.
11. String manipulation.
12. Libraries.
13. Data file reading and writing.
14. Source editors, compilers, and integrated debugging.

Course Objectives

1. Design and prepare programs in the Python programming language.
2. Create functions and modules.
3. Analyze problems for use of proper data types and logic structures.
4. Store and manage in-memory data with lists and dictionaries.

5. Utilize libraries to perform specialized processing.
6. Write and read text data files with Python.
7. Edit, compile, run and debug Python programs within an integrated set of software tools.

Method of Evaluation

1. Projects and hands-on labs: an example would include the writing of a program to solve a problem with a scientific or business situation or an interactive game.
2. Written tests and quizzes.
3. Final written or coding examination.

Special Materials Required of Student

1. Flash/USB drive or cloud storage for backup of in-class work.

Minimum Instructional Facilities

1. Standard computer lab with one internet-connected workstation per student with appropriate software installed.

Method of Instruction

1. Lecture
2. Practical and realistic assignments to apply concepts discussed.
3. Extensive practice in writing programs to run on computer. Simulation, where applicable, of real-life situations.

Out-of-Class Assignments

1. Prepare programming projects such as scientific, business, and action game programs in the designated language.
2. Textbook reading assignments.
3. Algorithm and problem-solving exercises

At least one text must be published within the last 5 years.

Representative Texts

	Author	Title	Publisher	Publication Year	ISBN
1	Lutz, Mark	Learning Python, 5th Edition	O'Reilly Media	2013	

Student Learning Outcomes

	Upon successful completion of this course, students should be able to:	Outcome Name
1	Given a problem specification, design and develop a Python program solution that successfully uses an ordered set of executable steps to successfully meet program requirements.	1
2	Compile, run, test and debug object-oriented Java programs using an integrated set of software tools.	2

Key: 3030